Course Code: B23CT4101

SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A) | **R23**

IV B.Tech. I Semester MODEL QUESTION PAPER

FUNDAMENTALS OF DEEP LEARNING

(Common to CSD & CSIT)

Time: 3 Hrs. **Max. Marks: 70 M**

Answer Question No.1 compulsorily

Answer **ONE Question** from **EACH UNIT**

Assume suitable data if necessary

10 X 2=20 Marks

			CO	KL	M
1	**a)**	Define **Deep Learning**.	**1**	**1**	**2**
	b)	What is meant by **model complexity** in deep learning?	**1**	**1**	**2**
	c)	What is a **Perceptron**?	**2**	**1**	**2**
	d)	What is the role of **bias input** in a perceptron?	**2**	**1**	**2**
	e)	Difference between the **batch training and sequential training**.	**3**	**2**	**2**
	f)	What is meant by **stochastic gradient descent (SGD)**?	**3**	**1**	**2**
	g)	Define a **Convolutional Neural Network (CNN)**.	**4**	**1**	**2**
	h)	Define **pooling** in convolutional networks.	**4**	**1**	**2**
	i)	Define a **Bidirectional RNN**.	**5**	**1**	**2**
	j)	What is meant by **explicit memory** in sequence modeling?	**5**	**1**	**2**
		5 x 10 = 50 Marks			
		UNIT-I			
2.	**a)**	Explain how **deep learning models achieve higher accuracy** compared to traditional machine learning techniques.	**1**	**2**	**5**
	b)	Describe the relationship between **complexity and performance** in deep learning models.	**1**	**2**	**5**
		OR			
3.	**a)**	Discuss the **real-world impact of Deep Learning** in various application domains.	**1**	**2**	**5**
	b)	Discuss the challenges and future scope of **Deep Learning technologies**.	**1**	**2**	**5**
		UNIT-II			
4.	**a)**	Explain the structure and working of the **biological neuron and artificial neuron**.	**2**	**2**	**5**
	b)	Explain the concept of **Linear Regression** and compare it with perceptron-based learning.	**2**	**2**	**5**
		OR			
5.	**a)**	Explain the **Perceptron Learning Algorithm** with a suitable example.	**2**	**2**	**5**
	b)	Define **Neural Networks** and explain their characteristics and	**2**	**2**	**5**

		applications.			
		UNIT-III			
6.	**a)**	Explain the architecture and working of a **Multi-Layer Perceptron (MLP)** with a neat diagram.	**3**	**2**	**5**
	b)	Describe the process of **initializing weights** in neural networks and its importance.	**3**	**2**	**5**
		OR			
7.	**a)**	Explain the importance of the **amount of training data and number of hidden layers** in MLP performance.	**3**	**2**	**5**
	b)	Discuss different criteria used to decide **when to stop learning** in neural network training.	**3**	**2**	**5**
		UNIT-IV			
8.	**a)**	Describe the **convolution operation** and explain how feature maps are generated.	**4**	**2**	**5**
	b)	Explain different types of **pooling techniques** used in CNNs.	**4**	**2**	**5**
		OR			
9.	**a)**	Discuss the role of **random or unsupervised features** in convolution networks.	**4**	**2**	**5**
	b)	Explain the **neuro scientific basis for convolution networks** and its influence on deep learning models.	**4**	**2**	**5**
		UNIT-V			
10.	**a)**	Explain the concept of **sequence modelling** and its importance in deep learning applications.	**5**	**2**	**5**
	b)	Explain the **Encoder–Decoder Sequence-to-Sequence architecture** and its applications.	**5**	**2**	**5**
		OR			
11.	**a)**	Discuss the **challenge of long-term dependencies** in RNNs and methods to overcome them.	**5**	**2**	**5**
	b)	Describe the architecture and advantages of **LSTM and other gated RNNs**. Explain the role of **explicit memory** in sequence modelling.	**5**	**2**	**5**

CO-COURSE OUTCOME **KL-KNOWLEDGE LEVEL** **M-MARK**

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

					Course Code: B23HS4101
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					**R23**
IV B.Tech. I Semester MODEL QUESTION PAPER					
HUMAN RESOURCE & PROJECT MANAGEMENT					
(Common to CSE, AIML, IT, AIDS, CSG, CSIT)					
Time: 3 Hrs.					**Max. Marks: 70 M**
Answer Question No.1 compulsorily					
Answer **ONE Question** from **EACH UNIT**					
Assume suitable data if necessary					
					10 x 2 = 20 Marks
			CO	**KL**	**M**
1.	**a).**	Write the role of HR manager.	**1**	**1**	**2**
	b).	Define recruitment.	**1**	**1**	**2**
	c).	List any two models of HR accounting.	**2**	**1**	**2**
	d).	Define career counseling.	**2**	**1**	**2**
	e).	Define Resource management.	**3**	**1**	**2**
	f).	State the 80-20 rule.	**3**	**1**	**2**
	g).	Define Brainstorming.	**4**	**1**	**2**
	h).	Write about Cash flow analysis.	**4**	**1**	**2**
	i).	List any two forms of project organization.	**5**	**1**	**2**
	j).	Define project control.	**5**	**1**	**2**
					5 x 10 = 50 Marks
		UNIT-1			
2.	**a).**	Interpret the nature and scope of Human Resource Management.	1	2	5
	b).	Discuss the emerging trends in HRM and their impact on organizations.	1	2	5
		OR			
3.	**a).**	Explain the ethical aspects of HRM with suitable examples.	1	2	5
	b).	Describe the steps involved in the selection procedure.	1	2	5
		UNIT-2			
4.	**a).**	Discuss the different methods of training used in organizations.	2	2	5
	b).	Explain the importance of performance appraisal in organizations.	2	2	5
		OR			
5.	**a).**	Summarize the concept of career development and its importance.	2	2	5
	b).	Explain group interaction and its importance in HRD.	2	2	5
		UNIT-3			
6.	**a).**	Explain the concept of Project Management and its importance.	3	2	5
	b).	Discuss the different types of projects with examples.	3	2	5

		OR			
7.	**a).**	Explain the stages of the project life cycle.	**3**	**2**	**5**
	b).	Interpret the process of project selection.	**3**	**2**	**5**
		UNIT-4			
8.	**a).**	Describe the criteria used for project selection.	**4**	**2**	**5**
	b).	A company invests ₹2,00,000 in a project. The expected cash inflows are: Year 1: ₹50,000 Year 2: ₹60,000 Year 3: ₹70,000 Year 4: ₹80,000 Calculate the Payback Period.	**4**	**2**	**5**
		OR			
9.		A project requires ₹2,00,000 investment. It generates ₹80,000 annually for 4 years. Discount rate = 10% Calculate the NPV and state whether the project is acceptable.	**4**	**2**	**10**
		UNIT-5			
10.	**a).**	Interpret the process of project planning in detail.	**5**	**2**	**5**
	b).	Explain the human aspects of project management.	**5**	**2**	**5**
		OR			
11.	**a).**	Discuss the prerequisites for successful project implementation.	**5**	**2**	**5**
	b).	Summarize the importance of project review.	**5**	**2**	**5**

CO-COURSEOUTCOME **KL-KNOWLEDGELEVEL** **M-MARKS**

NOTE: Questions can be given as A,B splits or as a single Question for10 marks

					Course Code: B23CS4102
		SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)			**R23**
		IV B.Tech. I Semester MODEL QUESTION PAPER			
		SOFTWARE ARCHITECTURE & DESIGN PATTERNS			
		(Common to CSE,AIDS, CSBS, CSD)			
Time: 3 Hrs.					**Max. Marks: 70 M**
		Answer Question No.1 compulsorily			
		Answer **ONE Question** from **EACH UNIT**			
		Assume suitable data if necessary			
					10 x 2 = 20 Marks
			CO	**KL**	**M**
1.	**a).**	Explain the role of Model–View–Controller (MVC) in design patterns.	**1**	**2**	**2**
	b).	Differentiate between cohesion and coupling**.**	**1**	**2**	**2**
	c).	Explain how actors are identified in use-case analysis.	**2**	**2**	**2**
	d).	Explain the concept of conceptual classes in system analysis.	**2**	**2**	**2**
	e).	Summarize the key idea behind the Adapter pattern.	**3**	**2**	**2**
	f).	Describe how the Proxy pattern controls access to another object.	**3**	**2**	**2**
	g).	Explain the role of the Model component in MVC architecture.	**4**	**2**	**2**
	h).	Explain how the undo operation can be supported in an interactive system.	**4**	**2**	**2**
	i).	Explain the concept of a distributed object system.	**5**	**2**	**2**
	j).	Explain the roles of client and server in a distributed application.	**5**	**2**	**2**
					5 x 10 = 50 Marks
		UNIT-1			
2.	**a).**	Describe the essential elements used to document a design pattern.	**1**	**2**	**5**
	b).	Describe how design patterns help solve common design problems in software development.	**1**	**2**	**5**
		OR			
3.	**a).**	Describe the key concepts of object-oriented design.	**1**	**2**	**5**
	b).	Describe the drawbacks of the object-oriented paradigm.	**1**	**2**	**5**
		UNIT-2			
4.	**a).**	Use requirement gathering techniques to prepare a functional requirement specification for a student attendance management system.	**2**	**3**	**5**
	b).	Demonstrate how use-case analysis can be used to represent the functional requirements of a bank ATM system.	**2**	**3**	**5**
		OR			
5.	**a).**	Illustrate the process of identifying conceptual classes and relationships from a given problem description.	**2**	**3**	**5**

	b).	Apply domain knowledge to refine the analysis model of an e-commerce order processing system.	**2**	**3**	**5**
		UNIT-3			
6.	**a).**	Demonstrate how the Bridge pattern can be used to separate device abstraction from its implementation in a remote-control system.	**3**	**3**	**5**
	b).	Illustrate the use of the Decorator pattern to dynamically add additional features to a text editor component.	**3**	**3**	**5**
		OR			
7.	**a).**	Apply the Facade pattern to simplify the interaction between multiple subsystems in a banking application.	**3**	**3**	**5**
	b).	Demonstrate how the Flyweight pattern can be used to optimize memory usage in a large document editor with many repeated characters.	**3**	**3**	**5**
		UNIT-4			
8.	**a).**	Apply the MVC architectural pattern to design an interactive drawing application, identifying the model, view, and controller components.	**4**	**3**	**5**
	b).	Apply MVC principles to design the subsystems of an interactive online shopping interface.	**4**	**3**	**5**
		OR			
9.	**a).**	Demonstrate how a new feature can be added to an MVC-based system without modifying the entire architecture.	**4**	**3**	**5**
	b).	Illustrate how pattern-based solutions improve extensibility in an MVC-based interactive system.	**4**		**5**
		UNIT-5			
10.	**a).**	Demonstrate how Java Remote Method Invocation (RMI) can be used to enable communication between distributed objects in a networked application.	**5**	**3**	**5**
	b).	Demonstrate how RESTful web services can be used to provide data access in a student information system.	**5**	**3**	**5**
		OR			
11.	**a).**	Illustrate the process of implementing an object-oriented system on the web for an online shopping application.	**5**	**3**	**5**
	b).	Illustrate how an Enterprise Service Bus (ESB) can be used to integrate different enterprise applications in a distributed environment.	**5**	**3**	**5**

CO-COURSEOUTCOME KL-KNOWLEDGE LEVEL M-MARKS

NOTE: Questions can be given as A,B splits or as a single Question for10 marks

Course Code: B23CD4101					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					**R23**
IV B.Tech. I Semester MODEL QUESTION PAPER					
INTRODUCTION TO BLOCKCHAIN TECHNOLOGY					
(For CSD)					
Time: 3 Hrs.					**Max. Marks: 70 M**
Answer Question No.1 compulsorily					
Answer **ONE Question** from **EACH UNIT**					
Assume suitable data if necessary					
10 X 2=20 Marks					
			CO	**KL**	**M**
1	**a)**	Define Mining.	**1**	**1**	**2**
	b)	List the types of Blockchain.	**1**	**1**	**2**
	c)	Define bitcoin.	**2**	**1**	**2**
	d)	List the cryptocurrency.	**2**	**1**	**2**
	e)	Compare Bitcoin and Ethereum.	**3**	**2**	**2**
	f)	Define Smart Contract.	**3**	**1**	**2**
	g)	Define Consortium Blockchain.	**4**	**1**	**2**
	h)	Define Hyperledger Platform.	**4**	**1**	**2**
	i)	List different types application of blockchain.	**5**	**1**	**2**
	j)	Define Self-Sovereign Identity	**5**	**1**	**2**
				1	
5 x 10 = 50 Marks					
		UNIT-I			
2.	**a)**	Explain the concept of decentralization and distributed ledger.	**1**	**2**	**5**
	b)	Describe the components of blockchain: Node, Ledger, Wallet, Hash, and Nonce.	**1**	**2**	**5**
		OR			
3.	**a)**	Explain the structure of a blockchain and how transactions are stored in blocks.	**1**	**3**	**5**
	b)	Explain briefly about the structure of blockchain and types of blockchain.	**1**	**2**	**5**
		UNIT-II			
4.	**a)**	Analyse the working of Bitcoin blockchain with neat diagram.	**2**	**4**	**5**
	b)	Explain the basics of cryptocurrency and its types.	**2**	**2**	**5**
		OR			
5.	**a)**	Describe PoW, PoS consensus algorithms.	**2**	**2**	**5**

	b)	Explain public blockchain and list popular public blockchains	**2**	**2**	**5**
		UNIT-III			
6.	**a)**	Explain the architecture of Ethereum briefly.	**3**	**2**	**5**
	b)	Explain the Merkle Patricia Tree and its role in Ethereum.	**3**	**3**	**5**
		OR			
7.	**a)**	Explain smart contracts and their characteristics.	**3**	**2**	**5**
	b)	Discuss the applications of smart contracts in industry.	**3**	**3**	**5**
		UNIT-IV			
8.	**a)**	Explain the key characteristics of consortium blockchain.	**4**	**2**	**5**
	b)	Explain the Hyperledger platform and its features.	**4**	**2**	**5**
		OR			
9.	**a)**	Discuss security and privacy challenges in blockchain.	**4**	**4**	**5**
	b)	Explain identity management and authentication in blockchain	**4**	**3**	**5**
		UNIT-V			
10.	**a)**	Explain the role of blockchain in banking and finance.	**5**	**3**	**5**
	b)	Discuss blockchain applications in healthcare and educatio	**5**	**3**	**5**
		OR			
11.	**a)**	Describe the integration of blockchain with IoT.	**5**	**4**	**5**
	b)	Discuss Self-Sovereign Identity and blockchain cloud offerings.	**5**	**3**	**5**

CO-COURSE OUTCOME **KL-KNOWLEDGE LEVEL** **M-MARKS**

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

					Course Code: B23CT4103
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					**R23**
IV B.Tech. I Semester MODEL QUESTION PAPER					
DATA SCIENCE USING PYTHON					
(Common to CSIT & CSD)					
Time: 3 Hrs.					**Max. Marks: 70 M**
Answer Question No.1 compulsorily					
Answer **ONE Question** from **EACH UNIT**					
Assume suitable data if necessary					
					10 X 2=20 Marks
			CO	**KL**	**M**
1	**a)**	State the concept of Data Science.	**1**	**1**	**2**
	b)	Outline the stages involved in the Data Science life cycle.	**1**	**1**	**2**
	c)	What is the purpose of feature selection.	**2**	**1**	**2**
	d)	Identify supervised learning with a suitable example.	**2**	**2**	**2**
	e)	Distinguish unsupervised learning from supervised learning.	**3**	**2**	**2**
	f)	Summarize Big Data and list any two characteristics.	**3**	**2**	**2**
	g)	State the role of Hadoop in Big Data processing.	**4**	**1**	**2**
	h)	Discuss the significance of NoSQL databases.	**4**	**2**	**2**
	i)	Explain the need for data preprocessing.	**5**	**2**	**2**
	j)	Illustrate the importance of data visualization in Data Science.	**5**	**3**	**2**
					5 x 10 = 50 Marks
		UNIT-I			
2.	**a)**	Explain the complete Data Science process with a neat diagram.	**1**	**2**	**10**
		OR			
3.	**a)**	Summarize the concepts and characteristics of Big Data.	**1**	**3**	**5**
	b)	Discuss the major challenges involved in Data Science.	**1**	**2**	**5**
		UNIT-II			
4.	**a)**	Explain the concept and significance of feature selection in Machine Learning.	**2**	**2**	**5**
	b)	Discuss different techniques used for handling missing data.	**2**	**2**	**5**
		OR			
5.	**a)**	Classify the types of Machine Learning and demonstrate each with suitable examples.	**2**	**3**	**10**
		UNIT-III			
6.	**a)**	Examine the Hadoop framework and explain the functions of its components.	**3**	**3**	**10**
		OR			

7.	**a)**	Interpret the working process of Map Reduce with a suitable example.	**3**	**3**	**5**
	b)	Demonstrate the advantages of NoSQL databases in Big Data applications.	**3**	**3**	**5**
		UNIT-IV			
8.	**a)**	Demonstrate various data visualization tools and techniques used in Python.	**4**	**3**	**10**
		OR			
9.	**a)**	Examine the features and applications of Mat plot lib or Sea born	**4**	**3**	**5**
	b)	Demonstrate any one real-time application of Data Science.	**4**	**3**	**5**
		UNIT-V			
10.	**a)**	Compare Artificial Intelligence, Machine Learning, and Deep Learning with suitable examples.	**5**	**4**	**10**
			5		**5**
		OR			
11.	**a)**	Examine the role and applications of Natural Language Processing (NLP).	**5**	**3**	**5**
	b)	Demonstrate any one real-time application of Artificial Intelligence such as chatbots or recommendation systems.	**5**	**3**	**5**

CO-COURSE OUTCOME **KL-KNOWLEDGE LEVEL** **M-MARKS**

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

Course Code:B23IT4103					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					**R23**
IV B.Tech. I Semester MODEL QUESTION PAPER					
COMPUTER VISION					
(Common to CSBS, CSE, IT, CSD)					
Time: 3 Hrs.					**Max. Marks: 70 M**
Answer Question No.1 compulsorily					
Answer **ONE** Question from **EACH UNIT**					
Assume suitable data if necessary					
					10 x 2 = 20 Marks
			CO	**KL**	**M**
1.	**a).**	Define radiometry and its importance in image formation.	1	1	2
	b).	What is meant by the local shading model? Give an example.	1	2	2
	c).	Define sampling and aliasing.	2	1	2
	d).	What is texture representation in images?	2	1	2
	e).	What is binocular fusion?	3	1	2
	f).	Define image segmentation.	3	1	2
	g).	What is the Hough Transform used for?	4	1	2
	h).	What is the purpose of the EM algorithm?	4	1	2
	i).	Define camera calibration.	5	1	2
	j).	What is radial distortion in cameras?	5	1	2
					5 x 10 = 50 Marks
		UNIT-1			
2.	**a).**	Compare different light sources and their effects on image formation and shadows.	1	3	5
	b).	Explain the working of a pinhole camera model and its limitations in real-world imaging.	1	2	5
		OR			
3.	**a).**	Describe photometric stereo and its role in recovering surface orientation	1	2	5
	b).	Explain local shading models, to interpret shading variations in an image.	1	2	5
		UNIT-2			
4.	a).	Explain Fourier Transform concepts in analyzing spatial frequency of images.	2	2	5
	b).	Discuss various edge detection techniques and their sensitivity to noise.	2	2	5

		OR			
5.	**a).**	Explain texture representation methods in texture analysis or synthesis.	2	2	5
	b).	Analyze the effects of sampling and aliasing in image processing with suitable examples.	2	3	5
		UNIT-3			
6.	**a).**	Explain stereopsis and how depth is reconstructed from two views.	3	2	5
	b).	Analyze the role of human visual perception (Gestalt principles) in grouping and segmentation.	3	3	5
		OR			
7.	**a).**	Compare graph-theoretic clustering methods for image segmentation.	3	3	5
	b).	Discuss the concept of image segmentation and clustering techniques for segmenting images.	3	2	5
		UNIT-4			
8.	**a).**	Explain data association in tracking and it in real-world scenarios.	4	2	5
	b).	Discuss model fitting as a probabilistic inference problem and its robustness.	4	2	5
		OR			
9.	**a).**	Analyze the working of Kalman filtering in tracking dynamic systems.	4	3	5
	b).	Explain the EM algorithm and it to solve missing data problems in segmentation.	4	2	5
		UNIT-5			
10.	**a).**	Discuss camera calibration techniques and least-squares estimation for parameter calculation.	5	2	5
	b).	Explain perspective projection and its role in camera modeling.	5	2	5
		OR			
11.	**a).**	Explain radial distortion and its methods to correct it.	5	2	5
	b).	Analyze the role of geometric camera models in applications like mobile robot localization or medical image registration.	5	3	5

CO-COURSEOUTCOME **KL-KNOWLEDGELEVEL** **M-MARKS**

NOTE: Questions can be given as A,B splits or as a single Question for10 marks

					Course Code: B23CD4102
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					**R23**
IV B.Tech. I Semester MODEL QUESTION PAPER					
ANIMATIONS PRINCIPALES DESIGN					
(For CSD)					
Time: 3 Hrs.					**Max. Marks: 70 M**
Answer Question No.1 compulsorily					
Answer **ONE Question** from **EACH UNIT**					
Assume suitable data if necessary					
					10 X 2=20 Marks
			CO	**KL**	**M**
1	**a).**	Define traditional animation.	**1**	**1**	**2**
	b).	Define 3D animation.	**1**	**1**	**2**
	c).	Where can anticipation be used in a character jumping action?	**2**	**1**	**2**
	d).	Describe staging in an animation scene.	**2**	**2**	**2**
	e).	State the meaning of unity in design.	**3**	**3**	**2**
	f).	Illustrate color application in creating mood.	**3**	**3**	**2**
	g).	Explain storytelling concepts in visual media.	**4**	**2**	**2**
	h).	Apply storyboarding techniques in visual planning.	**4**	**3**	**2**
	i).	Demonstrate the stages of animation production in a short project.	**5**	**3**	**2**
	j).	Demonstrate the application of AR technology in a simple animation-based application.	**5**	**3**	**2**
					5 x 10 = 50 Marks
		UNIT-I			
2.	**a).**	Explain the meaning of animation and describe how it has evolved over time from traditional methods to modern digital techniques.	**1**	**2**	**5**
	b).	Discuss the concept of traditional animation and explain how it is created using hand-drawn techniques	**1**	**2**	**5**
		OR			
3.	**a).**	Describe how animation is used in media and entertainment industries.	**1**	**2**	**5**
	b).	Explain the importance of animation in video games and interactive media	**1**	**2**	**5**
		UNIT-II			
4.	**a).**	Explain how the principle of squash and stretch enhances realism in character movement with suitable examples.	**2**	**2**	**5**
	b).	Illustrate the concept of staging and explain how clear presentation of action improves viewer understanding in animation scenes.	**2**	**3**	**5**
		OR			

5.	**a).**	Explain how anticipation and staging improve storytelling in animated films with suitable illustrations.	**2**	**2**	**5**
	b).	Illustrate the concept of solid drawing and explain how depth and volume are maintained in 2D animation.	**2**	**3**	**5**
		UNIT-III			
6.	**a).**	Examine the role of design elements such as line, shape, form, texture, and color in creating visually effective compositions.	**3**	**3**	**5**
	b).	Analyze how texture and color contribute to depth and realism in design works.	**3**	**4**	**5**
		OR			
7.	**a).**	Analyze the importance of color theory in creating visually appealing and meaningful designs.	**3**	**4**	**5**
	b).	Examine how color psychology influences audience perception in advertising and media design.	**3**	**4**	**5**
		UNIT-IV			
8.	**a).**	Evaluate the importance of narrative structure (Beginning, Middle, End) in storytelling with suitable examples.	**4**	**4**	**5**
	b).	Examine the role of storyboarding techniques in planning animation or film production.	**4**	**4**	**5**
		OR			
9.	**a).**	Illustrate how different camera angles and shots influence visual storytelling.	**4**	**3**	**5**
	b).	Differentiate script writing and storyboarding with respect to their purpose and application in media production.	**4**	**3**	**5**
		UNIT-V			
10.	**a).**	Explain the importance of pre-production, production, and post-production stages in animation.	**5**	**3**	**5**
	b).	Differentiate VR and AR technologies with suitable examples in animation.	**5**	**3**	**5**
		OR			
11.	**a).**	Compare 2D animation software and 3D animation software.	**5**	**4**	**5**
	b).	Examine the industry applications of animation in education, gaming, and advertising.	**5**	**4**	**5**

CO-COURSE OUTCOME **KL-KNOWLEDGE LEVEL** **M-MARKS**

NOTE: Questions can be given as A, B splits or as a single Question for 10 marks

Course Code: B23CD4104

SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A) | **R23**

IV B.Tech. I Semester MODEL QUESTION PAPER

INTRODUCTION TO AGILE METHODOLOGIES

(For CSD)

Time: 3 Hrs. | **Max. Marks: 70 M**

Answer Question No.1 compulsorily

Answer **ONE Question** from **EACH UNIT**

Assume suitable data if necessary

10 X 2=20 Marks

			CO	KL	M
1	a)	Define Software Engineering.	1	1	2
	b)	What is Software Crisis?	1	1	2
	c)	Define Requirement Engineering.	2	1	2
	d)	What are functional requirements?	2	1	2
	e)	Define software testing.	3	1	2
	f)	What is regression testing?	3	1	2
	g)	What is smoke testing?	4	1	2
	h)	List Agile Manifesto values.	4	1	2
	i)	Define User Story.	5	1	2
	j)	What are Story Points?	5	1	2

5 x 10 = 50 Marks

		UNIT-I			
2.	**a)**	Explain the evolution of Software Engineering.	1	2	5
	b)	Explain COCOMO estimation model.	1	2	5
		OR			
3.	**a)**	Estimate effort using LOC for a sample project.	1	2	5
	b)	Estimate effort using LOC for a sample project.	1	2	5
		UNIT-II			
4.	**a)**	Explain Requirement Engineering process.	2	2	5
	b)	Draw a Use Case Diagram for ATM system.	2	2	5
		OR			
5.	**a)**	Explain requirements elicitation techniques.	2	2	5
	b)	Create a Class Diagram for Student Management System.	2	2	5
		UNIT-III			

6.	**a)**	Compare verification and validation.	**3**	**3**	**5**
	b)	Analyze differences between smoke and sanity testing.	**3**	**3**	**5**
		OR			
7.	**a)**	Evaluate effectiveness of regression testing.	**3**	**3**	**5**
	b)	Apply boundary value analysis.	**3**	**2**	**5**
		UNIT-IV			
8.	**a)**	Explain Scrum Framework.	**4**	**2**	**5**
	b)	Describe Scrum roles and events.	**4**	**2**	**5**
		OR			
9.	**a)**	Explain XP practices.	**4**	**2**	**5**
	b)	Discuss Lean principles.	**4**	**2**	**5**
		UNIT-V			
10.	**a)**	Write user stories for online bookstore.	**5**	**2**	**5**
	b)	Estimate stories using Planning Poker.	**5**	**2**	**5**
		OR			
11.	**a)**	Analyze benefits and limitations of TDD.	**5**	**2**	**5**
	b)	Explain the activities involved in **Sprint Execution and Tracking**.	**5**	**2**	**5**

CO-COURSE OUTCOME KL-KNOWLEDGE LEVEL M-MARKS

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

					Course Code: B23CD4105
	SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)				**R23**
	IV B.Tech. I Semester MODEL QUESTION PAPER				
	GAME DESIGN AND DEVELOPMENT				
	(For CSD)				
Time: 3 Hrs.					**Max. Marks: 70 M**
	Answer Question No.1 compulsorily				
	Answer **ONE Question** from **EACH UNIT**				
	Assume suitable data if necessary				
					10 X 2=20 Marks
			CO	**KL**	**M**
1	**a)**	What are **serious games**? Give one example.	**1**	**2**	**2**
	b)	What is the purpose of a **Game Design Document (GDD)**?	**1**	**2**	**2**
	c)	What are **Vanden Berghe's Five Domains of Play**?	**2**	**2**	**2**
	d)	List any four **classic game genres**.	**2**	**2**	**2**
	e)	What is the relationship between a **player and an avatar**?	**3**	**2**	**2**
	f)	Mention any four elements involved in **character development**.	**3**	**2**	**2**
	g)	What is meant by a **nonlinear story**?	**4**	**2**	**2**
	h)	Define **granularity** in storytelling.	**4**	**2**	**2**
	i)	Define **Level Design** in video games.	**5**	**2**	**2**
	j)	What is meant by a **persistent world** in online gaming?	**5**	**2**	**2**
					5 x 10 = 50 Marks
		UNIT-I			
2.	**a)**	Discuss various types of **games for entertainment and serious games**.	**1**	**2**	**5**
	b)	Describe the **key components of video games** in detail.	**1**	**2**	**5**
		OR			
3.	**a)**	Explain the **anatomy of a game designer** and the challenges faced in game development.	**1**	**2**	**5**
	b)	Describe the **qualities and skills required for a successful game designer**.	**1**	**2**	**5**
		UNIT-II			
4.	**a)**	Discuss different **demographic categories of gamers** and their impact on game development.	**2**	**2**	**5**
	b)	Discuss various **devices used in gaming** and their advantages and limitations.	**2**	**2**	**5**
		OR			
5.	**a)**	Explain the concept of **game genre** and discuss its importance in game design.	**2**	**2**	**5**

	b)	Compare **home game consoles and personal computers** as gaming platforms.	**2**	**2**	**5**
		UNIT-III			
6.	**a)**	Discuss the concept of **creative and expressive play** in video games.	**3**	**2**	**5**
	b)	Explain the process of **getting an idea and converting it into a game concept**.	**3**	**2**	**5**
		OR			
7.	**a)**	Explain the role of **audio design** in enhancing the gaming experience.	**3**	**2**	**5**
	b)	Describe the **goals of character design** in video games.	**3**	**2**	**5**
		UNIT-IV			
8.	**a)**	Explain the **emotional limits of interactive stories** in games.	**4**	**2**	**5**
	b)	Discuss the advantages and disadvantages of **nonlinear storytelling** in games.	**4**	**2**	**5**
		OR			
9.	**a)**	Explain the importance of **storytelling in video games**.	**4**	**2**	**5**
	b)	Explain the factors to be considered while **writing stories for games** and discuss other important considerations in storytelling.	**4**	**2**	**5**
		UNIT-V			
10.	**a)**	Discuss the **advantages and disadvantages of online games**.	**5**	**2**	**5**
	b)	Explain the concepts of **persistent worlds and social problems** in online gaming.	**5**	**2**	**5**
		OR			
11.	**a)**	Explain the concept of **Level Design** and its importance in game development.	**5**	**2**	**5**
	b)	Discuss the common **pitfalls of level design** and how they can be avoided.	**5**	**2**	**5**

CO-COURSE OUTCOME **KL-KNOWLEDGE LEVEL** **M-MARKS**

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

			Course Code:B23IT4108		
		SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)			**R23**
		IV B.Tech. I Semester MODEL QUESTION PAPER			
		CYBER PHYSICAL SYSTEMS			
		(Common to IT, CSE, CSD)			
Time: 3 Hrs.			**Max. Marks: 70 M**		
		Answer Question No.1 compulsorily			
		Answer **ONE** Question from **EACH UNIT**			
		Assume suitable data if necessary			
					10 x 2 = 20 Marks
			CO	**K L**	**M**
1.	**a).**	Define Cyber Physical system	1	1	2
	b).	What are the Basic Techniques used in Cyber physical systems.	1	1	2
	c).	Define attack models in CPS security	2	1	2
	d).	Justify security importance in cyber-physical systems?	2	2	2
	e).	What is synchronization in distributed CPS?	3	1	2
	f).	Discuss fault tolerance challenges in distributed CPS	3	2	2
	g).	Differentiate between hard real-time and soft real-time systems	4	2	2
	h).	Explain the role of deadlines in scheduling	4	2	2
	i).	What is model integration in CPS?	5	1	2
	j).	Discuss multidisciplinary interactions in CPS design	5	2	2
					5 x 10 = 50 Marks
		UNIT-1			
2.	**a).**	Differentiate between traditional control systems and cyber-physical systems	1	2	5
	b).	Explain the motivation behind symbolic control design	1	2	5
		OR			
3.	**a).**	What are the challenges in controller synthesis for CPS?	1	1	5
	b).	Discuss safety and reachability specifications in CPS	1	2	5
		UNIT-2			
4.	a)	Explain the security challenges of Cyber-Physical Systems.	2	2	5
	b)	What are the major threats to CPS? Explain	2	1	5
		OR			
5.	**a).**	Explain redundancy as a countermeasure in CPS	2	2	5
	b).	Differentiate between insider and outsider attacks	2	2	5

		UNIT-3			
6.	**a).**	Discuss reliability and consistency issues in distributed CPS	3	2	5
	b).	Explain event-based synchronization methods	3	1	5
		OR			
7.	**a).**	Explain the role of formal software engineering in CPS	3	2	5
	b).	Explain the working of consensus algorithms	3	2	5
		UNIT-4			
8.	**a).**	Define Real-Time Systems and explain their importance in CPS	4	2	5
	b).	Explain fixed timing parameters in real-time scheduling	4	2	5
		OR			
9.	**a).**	Explain the role of memory in real-time scheduling	4	2	5
	b).	Explain variability and uncertainty in CPS scheduling	4	2	5
		UNIT-5			
10.	**a).**	Explain the motivation behind integrated modeling approaches	5	2	5
	b).	Discuss components and connectors in CyPhyML	5	2	5
		OR			
11.	**a).**	Discuss synchronous and asynchronous timing models.	5	2	5
	b).	Explain causal relationships in CPS models	5	2	5

CO-COURSE OUTCOME KL-KNOWLEDGE LEVEL M-MARKS

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

Course Code: B23CD4106

SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A) | **R23**

IV B.Tech. I Semester MODEL QUESTION PAPER

DIGITAL AUDIO DESIGN & SYNTHESIS

(For CSD)

Time: 3 Hrs. | **Max. Marks: 70 M**

Answer Question No.1 compulsorily

Answer **ONE Question** from **EACH UNIT**

Assume suitable data if necessary

10 X 2=20 Marks

			CO	**KL**	**M**
1	**a)**	Define amplitude and phase in sound.	1	1	2
	b)	State the Nyquist sampling theorem.	1	1	2
	c)	Differentiate between time-domain and frequency-domain representation.	2	2	2
	d)	What is a discrete-time audio signal.	2	1	2
	e)	Identify the purpose of equalization (EQ) in audio processing.	3	1	2
	f)	Explain the concept of dynamic range in audio signals.	3	2	2
	g)	Define sound synthesis.	4	1	2
	h)	Outline the basic concept of additive synthesis.	4	2	2
	i)	List the applications of digital audio in multimedia systems.	5	1	2
	j)	Explain the role of digital audio in speech processing.	5	2	2
			5 x 10 = 50 Marks		
		UNIT-I			
2.	**a)**	Explain the difference between analog audio and digital audio with suitable examples.	1	2	5
	b)	Illustrate the problem of aliasing in digital audio and explain how it can be avoided.	1	4	5
		OR			
3.	**a)**	Describe the PCM encoding process used in digital audio.	1	2	5
	b)	Differentiate WAV, MP3, and AAC audio file formats based on compression and usage.	1	2	5
		UNIT-II			
4.	**a)**	Illustrate the concept of Fourier Transform in audio signal processing.	2	2	5
	b)	Apply the concept of DFT to represent a discrete-time signal in frequency domain.	2	3	5
		OR			
5.	**a)**	Summarize the process of spectral analysis of audio signals.	2	2	5
	b)	Demonstrate the working principles of low-pass, high-pass, and band-pass filters with suitable examples.	2	3	5

		UNIT-III			
6.	**a)**	Describe the working of echo, delay, and reverb effects in audio systems.	3	2	5
	b)	Apply dynamic range processing techniques such as compression and limiting in audio signals.	3	3	5
		OR			
7.	**a)**	Classify different noise reduction techniques used in digital audio.	3	2	5
	b)	Demonstrate methods used for audio quality enhancement with suitable examples.	3	3	5
		UNIT-IV			
8.	**a)**	Explain the principle of subtractive synthesis with a suitable example.	4	2	5
	b)	Apply the concept of frequency modulation (FM) synthesis in generating complex sounds.	4	3	5
		OR			
9.	**a)**	Differentiate between additive and subtractive synthesis techniques.	4	2	5
	b)	Summarize the concept and applications of granular synthesis.	4	2	5
		UNIT-V			
10.	**a)**	Describe the use of digital audio in gaming and virtual reality environments.	5	2	5
	b)	Apply the concepts of digital audio in AI-based systems such as voice assistants.	5	3	5
		OR			
11.	**a)**	Interpret the working of audio processing in multimedia systems with suitable examples.	5	2	5
	b)	Illustrate real-world applications of digital audio through case studies such as Spotify or voice assistants.	5	2	5

CO-COURSE OUTCOME **KL-KNOWLEDGE LEVEL** **M-MARKS**

NOTE : Questions can be given as A, B splits or as a single Question for 10 marks

Course Code: B23CD4107

SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)	**R23**

IV B.Tech. I Semester MODEL QUESTION PAPER

BIG DATA ANALYTICS

(Common to CSD & CSIT)

Time: 3 Hrs. **Max. Marks: 70 M**

Answer Question No.1 compulsorily

Answer **ONE Question** from **EACH UNIT**

Assume suitable data if necessary

10 X 2=20 Marks

			CO	**KL**	**M**
1	**a)**	Define Big Data and list any four applications of Big Data Analytics.	**1**	**2**	**2**
	b)	Explain the purpose of wrapper classes in Hadoop serialization.	**1**	**2**	**2**
	c)	List the major differences between GFS and HDFS.	**2**	**2**	**2**
	d)	Explain the significance of replication factor in HDFS.	**2**	**2**	**2**
	e)	Describe the role of shuffle and sort phase in Map Reduce.	**3**	**2**	**2**
	f)	Explain the function of Record Reader in Map Reduce.	**3**	**2**	**2**
	g)	List the responsibilities of Mapper and Reducer with suitable examples.	**4**	**2**	**2**
	h)	Explain the working principle of FM Algorithm in stream processing.	**4**	**2**	**2**
	i)	List the features of Apache Spark	**5**	**2**	**2**
	j)	Compare Pig Latin and HiveQL.	**5**	**2**	**2**

5 x 10 = 50 Marks

		UNIT-I			
2.	**a)**	Explain the architecture of Hadoop Distributed File System (HDFS) and describe how it supports fault tolerance and scalability with a neat diagram.	**1**	**2**	**5**
	b)	Explain the concept of Serialization in Hadoop. Demonstrate how Writable interface is used for serialization with a suitable Java example.	**1**	**2**	**5**
		OR			
3.	**a)**	Explain the characteristics of Big Data. Discuss the challenges involved in storing and processing Big Data applications.	**1**	**2**	**5**
	b)	Explain Hadoop V1 and Hadoop V2 architectures and compare their building blocks with suitable diagrams.	**1**	**2**	**5**
		UNIT-II			
4.	**a)**	Explain different Hadoop cluster configurations namely Local mode, Pseudo-distributed mode, and Fully distributed mode with suitable examples.	**2**	**2**	**5**
	b)	Describe HDFS read and write operations with neat sketches and discuss the role of NameNode and DataNode during these operations.	**2**	**2**	**5**

		OR			
5.	**a)**	Discuss the anatomy of a Map Reduce job execution in YARN architecture with neat diagram.	**2**	**2**	**5**
	b)	Explain Rack Awareness in HDFS. How does Hadoop improve data reliability using replication and rack awareness?	**2**	**2**	**5**
		UNIT-III			
6.	**a)**	Explain the role of Combiner, Partitioner, and Record Reader in Hadoop Map Reduce framework with suitable examples.	**3**	**2**	**5**
	b)	Develop a Map Reduce program to calculate the frequency of words in a given input file and explain Mapper, Reducer, and Driver classes.	**3**	**3**	**5**
		OR			
7.	**a)**	Discuss Hadoop Streaming and explain how non-Java applications can be executed using Hadoop Streaming.	**3**	**2**	**5**
	b)	Implement Matrix Multiplication using Map Reduce and explain the processing steps involved in Mapper and Reducer phases.	**3**	**3**	**5**
		UNIT-IV			
8.	**a)**	Explain the Stream Data Model and Architecture. Discuss the importance of stream processing in real-time applications.	**4**	**2**	**5**
	b)	Illustrate the working of DGIM Algorithm for counting the number of 1's in a sliding window with a suitable example.	**4**	**3**	**5**
		OR			
9.	**a)**	Explain Apache Spark architecture and describe the functions of Spark Core, Spark SQL, Spark Streaming, MLlib, and GraphX.	**4**	**2**	**5**
	b)	Write a Spark program to demonstrate basic RDD transformations and actions. Explain the advantages of RDDs in Big Data processing.	**4**	**3**	**5**
		UNIT-V			
10.	**a)**	Explain Pig Architecture and discuss various data processing operators used in Pig Latin with suitable examples.	**5**	**2**	**5**
	b)	Illustrate the architecture of HBase and explain the role of ZooKeeper in coordinating distributed operations.	**5**	**3**	**5**
		OR			
11.	**a)**	Explain Hive Architecture and discuss different Hive data types with suitable examples.	**5**	**2**	**5**
	b)	Develop HiveQL queries to create a table, insert data, and perform filtering, grouping, and sorting operations with suitable examples.	**5**	**3**	**5**

CO-COURSE OUTCOME **KL-KNOWLEDGE LEVEL** **M-MARKS**

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks